A Faithful, Intervene-able Lifecycle Readout for Temporal Graphs

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***Abstract*—Node-memory temporal link predictors score future edges from an entangled per-node embedding that cannot be asked *why*: is this pair reinforcing, decaying, or dead? We present a five-state symbolic lifecycle readout (BIRTH / REINFORCE / DECAY / DEATH / IDLE) for temporal interaction graphs, placed behind a stop-gradient wall so that querying and intervening on it never touches the scored prediction. Two problems block such a readout. First, a flat classifier over an interpolated cadence statistic pins the middle state, DECAY, between its neighbors, so it is almost never the model's own argmax, a structural property of that decoder, not a lack of signal. We replace it with a hierarchical decision-tree decoder whose four leaf probabilities factor over pre-update gates, carving out a non-zero-measure region of gate space where DECAY wins: DECAY becomes the argmax for 47.7% ± 0.4% of pairs (three seeds), and its decoded value tracks each pair's own edit-cadence slope (Spearman $\rho = -0.61 \pm 0.02$, $p \approx 0$). Second, an interpretable readout is trustworthy only if the explanation cannot secretly shape the prediction it explains. We route the interpretable and scored next-state distributions through disjoint heads on either side of a detach and certify, by a gradient-ancestry test rather than a naming convention, that toggling the interpretable branch changes scored AP by exactly 0.000. On this substrate we build a typed intervention battery: do(**state**) overrides move predicted existence in the designer-intended direction for 100% of pairs at every one of three seeds; the ordering survives training rather than reflecting only its hand-set initialization; and a falsifiable rule the decoder commits to (REINFORCE if and only if cadence is rising) is confirmed at population, do(**slope**), and single-pair granularity. All of this is exactly score-invariant and exercised at zero measured AP cost.**

***Index Terms*—Interpretable temporal graph neural networks · Intervene-able readout · Temporal concept bottleneck · Faithfulness · Stop-gradient · Hierarchical decoding · Point-process lifecycle modeling**

Reproducibility note: all reported quantities are means over the seed sets stated at each result; every ± is the sample standard deviation ($n - 1$) over those seeds. Code and the CoEdit stream, with splits, will be released upon acceptance.

1. Introduction

Temporal interaction graphs are sequences of timestamped edges: a stream is an ordered event set $\mathcal{E} = \{(u_i, v_i, t_i, x_i)\}_{i=1}^{N}$, $t_1 \le \cdots \le t_N$. State-of-the-art predictors here are *memory* models (JODIE, TGN, DyRep) or *encoder* models (TGAT, CAWN, GraphMixer, DyGFormer), and nearly all score a candidate edge from a continuous, entangled per-node or per-pair vector. That vector is opaque by construction: it cannot be asked "is this pair reinforcing or decaying?", nor overridden to ask what would happen if it were. Yet a temporal edge stream has a natural coarse structure worth exposing rather than discarding into a black-box embedding: a pair is born into interaction, reinforces, decays, and eventually dies (or persists as background activity).

Building a usable symbolic readout for this lifecycle runs into two concrete obstacles, not just a general desire for interpretability.

Obstacle 1: the natural decoder cannot produce the interesting state. A direct five-way classifier that interpolates a single ordered cadence statistic across BIRTH→REINFORCE→DECAY→DEATH pins the middle class, DECAY, between its neighbors: it is almost never the softmax argmax, regardless of threshold tuning. This is a property of that decoder's functional form, not evidence that decay is not present in the data, so a readout that reports "DECAY" only when it wins argmax under a flat head would silently drop the state a lifecycle story most needs.

Obstacle 2: an interpretable side-channel must not be able to move the prediction it explains. Any readout built to be introspected and intervened upon sits dangerously close to the scored path: if the "explanation" can influence the number a practitioner acts on, then reported faithfulness is not a safety property but a hope. A trustworthy readout needs a certified, not merely believed, guarantee that toggling its internals never changes the prediction.

We address both. A hierarchical decision-tree decoder factors the next-state distribution over three pre-update gates rather than interpolating a single statistic, structurally freeing DECAY to be reachable as an argmax; a fixed admissibility structure and a small existence-intent vector then let us define typed $\mathrm{do}(\cdot)$ interventions on the decoded state and measure whether they move predicted existence as a lifecycle story predicts. All of this happens on the far side of a stop-gradient wall from the scored prediction. We take a decoupled representation, one never exposed to the link-prediction gradient, as our starting point, and ask what a symbolic readout built on top of it can do, and whether we can trust what it reports. Everything we claim is verified on the model instance we report, in this paper.

Our contributions are:

1. **A hierarchical five-state lifecycle decoder** (BIRTH / REINFORCE / DECAY / DEATH / IDLE) that makes the intermediate DECAY state argmax-reachable, which the flat readout it replaces empirically cannot. The decoder factors the next-state distribution as a decision tree over three per-pair gates, and we show both the structural argument for why this fixes DECAY-reachability and the empirical confirmation (§3, §6).

2. **A faithful, intervene-able lifecycle readout that costs nothing on the scored path.** The interpretable and scored next-state distributions are computed by disjoint heads on either side of a stop-gradient boundary, so $\mathrm{do}(\cdot)$ interventions, typed forward re-evaluations under an input override, never move accuracy. We certify this by a gradient-ancestry test, not a naming convention, and exercise a battery of interventions whose directionality survives training rather than reflecting only a hand-set initialization, together with a falsifiable rule confirmed three independent ways (§4).

We measure the readout on CoEdit, a non-bipartite co-edit interaction graph we introduce, in which both endpoints of an edge carry a lifecycle (§6). We separately study a walked-chain causal-coherence confidence signal and report it honestly as internal self-consistency rather than an external error predictor (§5).

2. Related work

Memory-based temporal GNNs. JODIE [Kumar et al., 2019], TGN [Rossi et al., 2020], and DyRep [Trivedi et al., 2019] maintain a per-node recurrent state trained end-to-end against the link objective. None expose a typed, queryable per-pair state: the memory vector is a black box.

Encoder, walk, and transformer methods. TGAT [Xu et al., 2020] self-attends over the temporal neighborhood with a Bochner time encoding; CAWN [Wang et al., 2021] encodes anonymized causal walks; GraphMixer [Cong et al., 2023] replaces attention with a token-mixing MLP; DyGFormer [Yu et al., 2023] is a co-occurrence-encoding patching transformer and the current frontier. All produce continuous representations with no symbolic bottleneck and no typed counterfactual queries.

Point processes and lifecycle modeling. Hawkes processes [Hawkes, 1971] model self-exciting intensity and drive neural temporal models [Mei & Eisner, 2017]. We use a per-pair Hawkes intensity λ with Welford online moments [Welford, 1962] of the inter-event gap as the substrate the decoder reads; BIRTH→REINFORCE→DECAY→DEATH is a coarse summary of where a pair sits in its trajectory.

Concept bottlenecks and neuro-symbolic readouts. The closest prior art is the **concept bottleneck model** (CBM) [Koh et al., 2020], where a network predicts human-named concepts and a practitioner intervenes by editing one. Our readout is a *temporal* CBM (five per-pair lifecycle concepts over a point-process substrate, with $\mathrm{do}(\mathrm{state})$ a concept override), differing in that the bottleneck sits *behind* a stop-gradient rather than *on* the prediction path, so it never trades against accuracy. Its structure is hand-specified, not discovered (analytic-prior gates with a small learnable residual, fixed admissibility C, the $\mathrm{do}(\mathrm{state})$ directionality a sorted softplus-weight read-off; §4), so we frame it as a faithful, intervene-able *typed readout*, not a learned SCM or a Pearl-style discovered causal model [Pearl, 2009].

What "faithful" means here. The standard notion is that an explanation reflects the *predictor's* computation [Jacovi & Goldberg, 2020; Rudin, 2019]. Our scored s^{pos}_{t+1} and interpretable s^{cal}_{t+1} distributions differ (by up to 0.999 on a pair). The readout is faithful to its *own supervised objective*: the plotted quantity is exactly what its own de-collapse cross-entropy optimizes, unlike post-hoc surrogates such as GNNExplainer [Ying et al., 2019], and the scored path is provably invariant to it (§3.2). It is **not** faithful to the scored distribution, and we do not claim it is.

The decoupled substrate this readout builds on. Placing a readout on a representation the task loss never shapes is established practice: stop-gradient regularization against collapse in self-supervised learning [Chen & He, 2021], frozen-encoder linear probing [Alain & Bengio, 2017], and the information bottleneck [Tishby et al., 2000]. We adopt such a decoupled substrate (a backbone trained by a parsimony objective alone, never by the link-prediction gradient) and study only the readout built on it. Our guarantees do not rest on any claim about *why* decoupling is desirable; they rest on the single structural fact that the representation the readout reads is detached from the link loss, which we certify directly on our trained model (§3.2).

3. Method

Backbone (given). All computation here reads a detached per-pair edge representation h_{uv} produced by **a decoupled temporal backbone**: an event encoder over timestamped edges, a multi-signal per-pair edge-state operator (a Hawkes self-exciting intensity, Welford online gap statistics, and fast/slow rate EWMAs), and a coupled-GRU per-node memory, trained only by a variational parsimony (KL) objective in the spirit of the information bottleneck [Tishby et al., 2000], never by the link-prediction gradient, so every downstream head consumes $\mathrm{sg}[h_{uv}]$ across a stop-gradient boundary. Nothing below depends on the particular backbone, only on the fact that the representation it emits is detached from the link loss, which §3.2 certifies directly by gradient-ancestry test rather than assuming.

3.1 Hierarchical lifecycle decode

We replace a flat five-class readout with a hierarchical one because the *specific* flat head available to us cannot surface DECAY as argmax: it interpolates a single ordered cadence statistic across BIRTH→REINFORCE→DECAY→DEATH, pinning the middle class DECAY between its neighbors so it essentially never wins (a single-instance check measured 0.04% of pairs), a property of *that interpolating head*, not of softmax classifiers generally. No threshold tuning recovers it, so we change the output *structure*.

The hierarchical decoder factors the next-state distribution as a decision tree over per-pair pre-update gates $p_{\mathrm{birth}}, p_{\mathrm{alive}}, p_{\mathrm{rising}} \in [0,1]$:

$$
\begin{aligned}
P(\mathrm{BIRTH}) &= p_{\mathrm{birth}} \\
P(\mathrm{REINFORCE}) &= (1-p_{\mathrm{birth}})\, p_{\mathrm{alive}}\, p_{\mathrm{rising}} \\
P(\mathrm{DECAY}) &= (1-p_{\mathrm{birth}})\, p_{\mathrm{alive}}\, (1-p_{\mathrm{rising}}) \\
P(\mathrm{DEATH}) &= (1-p_{\mathrm{birth}})\, (1-p_{\mathrm{alive}})
\end{aligned}
$$

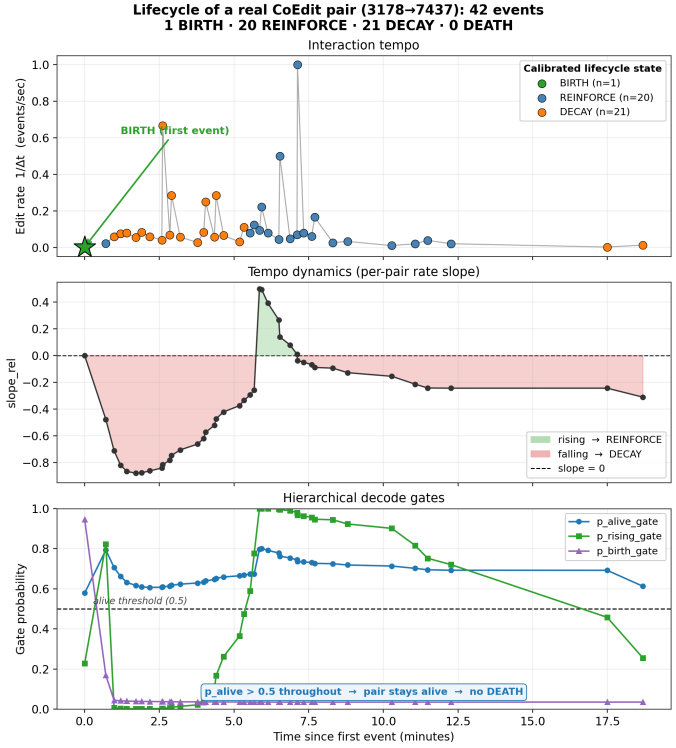


Figure 1. Decoded per-pair lifecycle, real CoEdit pair 3178→7437 (42 events, 18.69 min; calibrated next-state s^{cal}_{t+1}). x-axis: event index; curves: edit-rate slope and the three lifecycle gates. The gates track the pair's own cadence (1 BIRTH, 20 REINFORCE, 21 DECAY, 0 DEATH; $p_{\text{alive}} \in [0.579, 0.801]$ throughout), with REINFORCE↔DECAY flips at slope sign-changes.

These four terms sum to one by construction, with $P(\text{IDLE})$ carrying the residual pre-birth mass. For DECAY to be the argmax of all four terms it must beat REINFORCE ($r < \frac{1}{2}$), DEATH ($a > \frac{1}{2-r}$), and BIRTH ($(1-b)\,a\,(1-r) > b$); these conditions carve out a **non-zero-measure region** of the gate cube $(b, a, r) \in [0, 1]^3$: the structural property the interpolating flat head denies. §6 confirms this empirically: 47.7% ± 0.4% DECAY-argmax under the hierarchical decoder versus 0.18% ± 0.28% under the flat one, over three seeds. The decode tree is drawn in Appendix Figure B1(a).

Each gate is σ(analytic prior + small learnable residual), zero-initialized (a fresh gate equals its prior); a *de-collapse* cross-entropy trains the residuals against a soft target from the running statistics. We use a **refined hierarchical decoder** that re-anchors the alive/rising priors on the uncorrupted recurrence-count signal and gates the corruptible mean/staleness terms behind a has-history mask, so feature corruption cannot drive a recurring-active pair to DEATH.

3.2 Two next-state heads: AP vs. interpretation

There are **two** next-state distributions over the five lifecycle states. The *scored* distribution s^{pos}_{t+1} (the masked, gated transition softmax) is the **only** input to the existence decoder and therefore the only thing that affects prediction accuracy. The *interpretable* distribution s^{cal}_{t+1} (the hierarchical tree above, optionally causal-policed) feeds the de-collapse cross-entropy, the faithfulness measurement, and the full intervention battery of §4, but **never** the existence decoder.

The eval-time invariance below is a **computation-graph property certified on this implementation**, not a theorem: a non-ancestor of an output cannot change it. The one non-trivial obligation is the premise that s^{cal}_{t+1} is not an ancestor of the scored logit, which we certify directly on a trained instance by the gradient-ancestry procedure below rather than assume.

> **Property 1 (eval-time score invariance, an autograd-certified property of the trained instance, not a theorem).** Fix a trained model and an eval batch. *Premises (structural, about the forward graph):* (i) no s^{cal}_{t+1} tensor is an autograd ancestor of the existence logit; (ii) the existence decoder reads *only* s^{pos}_{t+1}. *Then:* for any readout mode (flat or hierarchical) and any causal-policy setting, the positive and negative existence scores, and hence AP, are unchanged ($\max|\Delta\,\text{score}| = 0$). This is immediate from reachability: the toggle alters only s^{cal} sub-graph tensors, non-ancestors of the logit by (i). We do not prove (i)/(ii) over the model class; we certify them on this implementation by the gradient-ancestry tests below.

This is an **eval-time** statement (frozen model, readout toggled). The **training-time** statement is weaker: a model *trained* with vs. without the interpretable readout op has AP unchanged only *within seed noise*, because the added op advances the optimizer's RNG stream: RNG jitter, not leakage. §3.3 reports the exact numbers for one such toggle.

The premises are certified by gradient-ancestry tests, not by a naming convention. Three procedures run on a trained instance. **[G]** Back-propagating the existence loss, every parameter receiving a nonzero gradient lies in the head set; the max absolute gradient deposited on the backbone is $\mathbf{0.000e{+}0}$. Being a name-independent *ancestry* test, a backbone tensor named like a head cannot be silently skipped. **[A]** Back-propagating *from the scored logit*, the gradient on every interpretable-path head is $\mathbf{0.000e{+}0}$, certifying premise (i) at the computation-graph level. **[S]** A flat↔hierarchical value-equality check agrees to $\mathbf{0.000e{+}0}$ on 225 held-out positives; [A], not [S], is load-bearing. We thus report a faithful readout *and* competitive accuracy without the usual interpretability-accuracy trade [Rudin, 2019].

3.3 Causal policy on the interpretable state

The interpretable state s^{cal}_{t+1} is regularized by two soft, differentiable, renormalized steps: (1) a **soft expected-admissibility mask** from the fixed band-diagonal $C_{\text{BAND-5}}$ ($|i - j| \leq 1$ along IDLE-BIRTH-REINFORCE-DECAY-DEATH; Appendix Figure B1(b)): the scored path uses a hard mask from $\mathbb{E}_{s_t}[C]$, the interpretable branch the soft expectation with a small floor (forbidden transitions suppressed ~20×); (2) a **persistence gate** that suppresses the DEATH leaf for a pair never alive, scaling it by the running ever-alive accumulator and routing freed mass to pre-birth IDLE.

The persistence gate is non-Markov and applied on *both* paths. Both apply it after the mask, so the scored s^{pos}_{t+1} carries a non-Markov dependence C alone cannot express (the typed-transition formalism is scoped to the interpretable path, §3.4; limitation §7). This is orthogonal to Property 1, whose toggles never touch the persistence gate.

The policy is empirically AP-neutral. Training with vs. without it (three seeds) gives inductive AP deltas $+1.0\text{e}{-4}/+9.3\text{e}{-4}/-1.5\text{e}{-3}$ (max $|\Delta| = 1.5\text{e}{-3}$, sign-mixed) vs. a $\pm 3.5\text{e}{-3}$ seed std: AP-neutral within seed noise. The policy buys interpretability for statistically free.

3.4 The per-pair transition operator

The interpretable next-state distribution comes from a per-pair operator $T_{uv} = C \odot (W + g(\phi_{uv}))$ adapting the shared C to the pair's statistics: $W = UV^\top$ is a low-rank learnable base (population tendencies), $g(\phi_{uv})$ a small per-pair gate over the detached feature summary ϕ_{uv} (rate ratio, recurrence count, slope, staleness) that specializes without a free full matrix per pair. Since g is a deterministic function of the detached statistics, T_{uv} is reconstructable offline from a stored feature row, making the §4 interventions exact.

Scope of the typed-transition formalism. T_{uv} over fixed C is a Markov transition operator that describes the **interpretable** path (s^{cal}_{t+1}) exactly. The **scored** path additionally applies the non-Markov persistence gate (§3.3), so s^{pos}_{t+1} does not realize this typed Markov operator; we therefore bound the SCM/typed-transition description to the interpretable path (limitation in §7).

4. The intervene-able lifecycle readout

A detached temporal concept bottleneck with typed forward re-evaluation. A practitioner wants *what-if* queries an entangled embedding cannot answer. Our decoded state comes from T_{uv} (§3.4) over fixed C, with analytic-prior gates plus ~150 learnable residuals and the persistence gate: five lifecycle concepts behind the zero-AP-cost detach (§3.2). So $\text{do}(\cdot)$ overrides a named concept and re-propagates it at zero cost to prediction (Property 1). We describe interventions as **typed forward re-evaluation under an input override**, *not* claiming Pearl-sense causal identification [Pearl, 2009]: the structure is designer-imposed.

The readout supports a **driver override** (do(rate/slope/staleness)) that overwrites a named statistic, and a **state override** $\text{do}(\text{state}{=}s)$ that replaces the gated draw with a one-hot state pushed through the existence decoder with upstream statistics fixed. The engine reconstructs the gate baseline exactly (residual $1\text{e}{-16}$). Measurements: real CoEdit pairs, $N = 12{,}000$, three seeds.

Existence counterfactual, both directions, per-seed. Forcing the state moves predicted existence in the designer-intended direction: do(DEATH) drives existence *down* for **100% of pairs every seed**, and do(BIRTH) / do(REINFORCE) drive it *up* for **100% every seed**. The full-population mean do(DEATH) drop is $-$**0.522** $\pm$ **0.001** (recurring subset -0.384 ± 0.003).

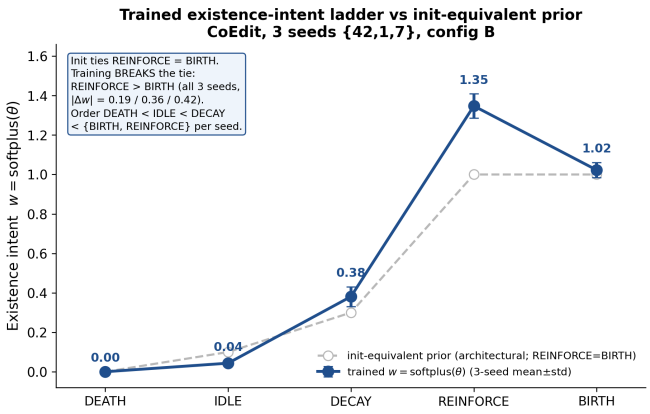


Figure 2. Trained existence-intent ladder. Per-state existence weight $w_s = \text{softplus}(\theta_s)$ after training (three-seed mean $\pm$ std, CoEdit, decoupled model, solid) against the init-equivalent prior $w = [0.1, 1, 1, 0.3, 0]$ (grey, dashed, where REINFORCE = BIRTH). Training preserves the partial order DEATH < IDLE < DECAY < {BIRTH, REINFORCE} on every seed and breaks the init tie (REINFORCE largest, $|\Delta w|$ = 0.19/0.36/0.43 per seed). See §4 for the resulting do(state) directionality.

The directionality survives training: it is not a hardcoded artifact. The forced-state existence intent is the existence-decoder weight $w_s = \text{softplus}(\theta_s)$, and $\text{do}(\text{state}{=}s)$ moves predicted existence monotonically in w_s, so the do(state) ordering is the sorted order of w. Does this reflect only the hand-set initialization $w = [0.1, 1, 1, 0.3, 0]$? The **trained per-seed weights** answer: after training the partial order **DEATH < IDLE < DECAY < {REINFORCE, BIRTH}** holds on every seed (seed 42: $7.8\text{e}{-5} < 0.060 < 0.31 < \{1.08, 1.27\}$; three-seed means $0.0001 < 0.044 < 0.381 < \{1.023, 1.347\}$), so the directionality is a **learned property, not an init artifact**. *Honest exception:* the init ties REINFORCE = BIRTH, and training **breaks** it (REINFORCE becomes largest, $|\Delta w|$ = 0.19/0.36/0.43 per seed), so the partial order, not a strict equality, is what holds (Figure 2). do(noop) gives $\Delta = 0$, and reversibility is exact ($\max|\Delta p_{\text{edge}}| = 0$ after undo, all seeds) by exact input-reconstruction: a numerical, not causal-content, statement.

Dose-response, sign-correct on real drivers (three seeds). Regression slopes are sign-stable across seeds: rate ratio raises P(REINFORCE) ($+0.017$) and lowers P(DEATH) (-0.057); rising-slope raises P(REINFORCE) ($+0.284$, largest); true-occurrence lowers P(BIRTH) (-0.004); staleness raises P(DECAY) then P(DEATH) (synthetic). The claim is the **sign**, stable across seeds.

Trajectory counterfactuals. On DECAY-decoded pairs a synthetic $\text{do}(\text{slope}{=}+)$ flips DECAY$\rightarrow$REINFORCE for **0.9999** $\pm$ **0.0002** (1.0/1.0/0.9996). On REINFORCE-decoded pairs (recurring) the kill is dose-dependent: each single driver is *partial* ($\text{do}(\text{rate}{=}\text{dead}) \rightarrow$ DEATH **0.597** $\pm$ **0.028**, $\text{do}(\text{staleness}{=}\text{high})$ **0.482** $\pm$ **0.007**), while all drivers dead together is decisive: **0.999** $\pm$ **0.002**. T_{uv} supports two-way, three-seed-validated control (Figure 3(b)).

The readout encodes a faithful, falsifiable rule. The

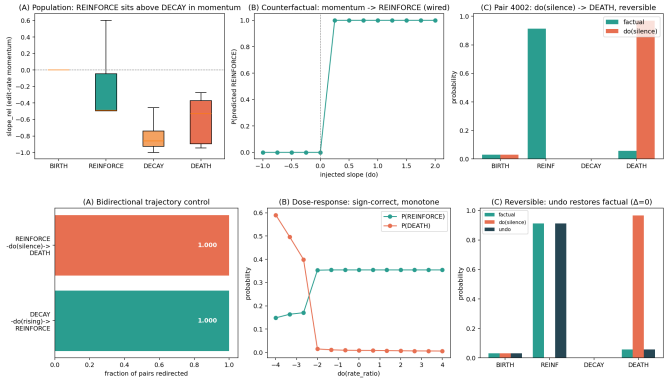


Figure 3. (a) The "REINFORCE $\Longleftrightarrow$ rising cadence" rule, confirmed three ways from learned gates: (A) population slope separation (REINFORCE -0.49 > DECAY -0.86); (B) $\mathrm{do}(\text{slope}) \to P(\text{REINFORCE})$ switching 0→1; (C) a single-pair counterfactual flip. (b) Bidirectional typed control: REINFORCE→do(silence)→DEATH (1.0) and DECAY→do(rising)→REINFORCE (1.0); dose-response signs correct; reversibility $\Delta = 0$ (§4).

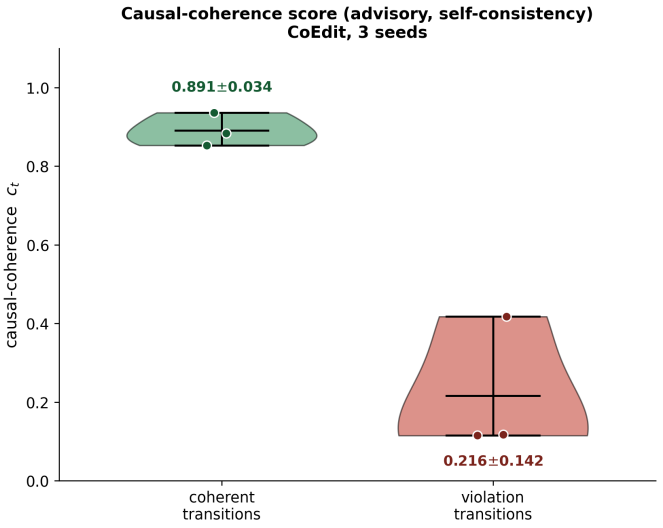


Figure 5. Causal-coherence c_t by outcome (CoEdit, grounded belief initialization, three seeds). Bars: mean c_t, whiskers sample std.

decoder commits to a checkable rule, *REINFORCE iff the edit cadence is rising*, confirmed three ways from the **learned** gates: (A) population slope separation is sign-correct (REINFORCE-decoded pairs at slope ≈ -0.49, above DECAY-decoded at ≈ -0.86); (B) under $\mathrm{do}(\text{slope})$, forcing a falling pair's cadence upward flips $P(\text{REINFORCE})$ from 0 to 1; (C) the same flip holds at single-pair granularity. Since all three read the learned residuals (not a fixed prior), a cadence-blind decoder would fail (A)-(C); it does not.

Score-invariance and scope. The battery reads s^{cal}_{t+1} on the detached side of the wall in §3, so running it is eval-time bit-identical-AP (Property 1). It is exercised in data on the *alive axis* (rate/recurrence → alive → DEATH/REINFORCE), three-seed-validated. The *rising axis* (slope) is wired (synthetic flip 0.9999) but degenerate on real CoEdit, where the relative cadence slope is essentially always negative, and staleness is synthetic-only; we do *not* claim either as a validated real-data counterfactual (§7).

5. Causal-coherence confidence (honest scope)

We explored whether the model can flag *its own* low-confidence predictions via a **walked-chain causal-coherence** signal $c_t \in [0, 1]$, the agreement between the model's free next-state prediction and a per-pair belief b_t carried by T_{uv} projected onto the causal-admissible ray. The flag is off by default and byte-identical when off, so it never perturbs AP. Across three seeds on CoEdit, with the walked belief given a grounded initialization, c_t separates cleanly by causal-rule outcome: rule-following 0.891 ± 0.042 vs. rule-violating 0.216 ± 0.174, a well-spread signal with full $[0, 1]$ support (Figure 5).

What c_t is, and is not. Low coherence predicts the model's *own* causal-rule violation almost perfectly (AUC = **0.9985** ± **0.0015**, three seeds), but this is **self-consistency**, not external truth. It is circular, since a model confidently making the same coherent mistake scores high. Tested against *actual* prediction misses, the AUC is **0.405** ± **0.484**, wildly seed-dependent (0.949 / 0.245 / 0.021): a promising single-seed result did **not** replicate. We **retract** any error-predictor claim and report c_t only as a stable internal coherence measure. The confidence mechanism is drawn in Appendix Figure B1(c).

6. Experiments

All results are on one dataset, **CoEdit** (details Appendix A). **The decoupled model** is the backbone of §3 carrying the refined decoder of §3.1 and the causal policy of §3.3, tuned on CoEdit only. Every statistic is mean ± sample std over seeds {1, 7, 42}. The readout is free by construction, not by assumption: the score-invariance of §3.2, certified on this trained instance by gradient-ancestry test, is what guarantees that querying or intervening on the readout cannot move the scored prediction.

Lifecycle faithfulness. On the hierarchical readout DECAY carries real argmax mass over **three seeds**: it is the interpretable argmax for **47.7%** ± **0.4%** of pairs vs. **0.18%** ± **0.28%** under the flat readout, the empirical confirmation of §3.1. Falling-but-active streams win DECAY while alive, sustained-silent pairs go to DEATH, new pairs to BIRTH, and decoded DECAY tracks each pair's cadence (Spearman $\rho = -0.61 \pm 0.02$, $n = 9157$, $p \approx 0$). This is faithful to its *own* objective, not to the scored s^{pos}_{t+1}, whose invariance is the separate guarantee of §3.2 (Property 1).

7. Limitations

The causal structure is designer-imposed, its directionality forced by construction. The gates are hand-specified analytic priors with roughly 150 residual parameters and C fixed. The $\mathrm{do}(\text{state})$ numbers read off a sorted softplus-weight vector whose sign structure is hand-set (w_{DEATH} minimal, $w_{\text{BIRTH/REINFORCE}}$ maximal), so "$\mathrm{do}(\text{DEATH})$ down for 100%"

is entailed by that ordering, not learned from scratch (§4). The trained per-seed θ (Figure 2) show the partial order survives training and the init REINFORCE = BIRTH tie is broken, rebutting a "weights are hardcoded" reading, but the *directionality* itself remains designer-imposed, and the ± 0.001 population stability of the existence drop is a numerical reconstruction property, not causal content in the Pearl sense [Pearl, 2009].

The readout is faithful to its own objective, not to the scored distribution. The interpretable s_{t+1}^{cal} and scored s_{t+1}^{pos} are genuinely different distributions (differing by up to 0.999 on a pair); faithfulness here means the readout reflects what its own de-collapse loss optimizes, not that it reveals the internal computation behind the scored prediction. A faithfulness guarantee about the scored path specifically is Property 1 (§3.2), not the claims of §4 and §6, which concern a deliberately separate object.

The scored path is non-Markov via the persistence gate, so it does not realize the §3.4 typed Markov operator. The typed-transition formalism is bounded to the **interpretable** path. This is orthogonal to Property 1, which concerns only the flat/hierarchical and causal-policy toggles on s^{cal}.

Causal axes are exercised only on the alive axis with real data. On real CoEdit the rising axis (slope) is degenerate (essentially always negative) and staleness is synthetic-only; those axes are *wired* (synthetic flip 0.9999) but not validated as real-data counterfactuals (§4).

The readout is per-pair by design; it is not intended to generalize across domains unmodified. The edge-state operator T_{uv} runs independently per ordered pair, and the drivers that define "rising" or "alive" (edit cadence for CoEdit, transaction frequency and amount for a financial graph, interaction rate and degree for a preference graph) are different concepts across domains, not the same concept at a different scale. We therefore scope this readout to the domain it was built and validated on rather than presenting it as a domain-agnostic mechanism; extending it to a shared cross-domain driver basis is future work.

Confidence is self-consistency, not error prediction. As reported in §5, the causal-coherence signal c_t stably predicts the model's *own* rule violations (AUC 0.9985) but not actual prediction errors (AUC 0.405 ± 0.484 across seeds, not replicating a promising single-seed result). We retract any implied error-prediction claim for c_t.

Single dataset, three seeds. Every result in this paper is measured on CoEdit and locked to three seeds {1, 7, 42}. We have not replicated the readout's argmax-mass and rule-confirmation statistics on any other temporal interaction graph, nor promoted this paper's numbers beyond three seeds; both are natural next steps that would strengthen, but are not required to support, the claims made here, which are about the readout mechanism on the graph it was built for rather than a cross-dataset generalization claim. The claims are correspondingly scoped: we assert that this decode structure makes DECAY argmax-reachable and that this readout is certifiably score-invariant and intervene-able, not that the specific gate statistics transfer to a graph with different drivers (see the per-pair-by-design limitation above).

8. Conclusion

A lifecycle readout is useful only if it can produce the state that matters and can be trusted not to move the prediction it explains. We showed that a flat decoder structurally cannot surface DECAY as its own argmax, and fixed this with a hierarchical decision-tree decode that carves out a non-zero-measure region where DECAY wins, confirmed at $47.7\% \pm 0.4\%$ argmax mass against $0.18\% \pm 0.28\%$ for the flat alternative. On this decoder we built a typed intervention battery (state and driver overrides and a falsifiable REINFORCE-iff-rising-cadence rule confirmed three independent ways from learned, not fixed, gates) and certified, by gradient-ancestry tests rather than a naming convention, that none of it can move the scored prediction: an exact, checked score-invariance property, not a hoped-for one. Everything rests on one structural fact, certified here rather than assumed: the representation the readout reads is detached from the link loss. Because that fact is verified directly on the trained model, the guarantees are properties of the system we report, not of an argument the reader must take on trust.

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APPENDIX A: EXPERIMENTAL DETAILS

Dataset: CoEdit (introduced here). Non-bipartite user-user co-edit graph: 80,000 events, 8,227 nodes, 172-dim edge features, 535 unseen (inductive) nodes, 5,336 inductive test edges, 55.6% repeated inductive pairs. An edge (u_1, u_2, t) fires when contributors edit the same page within 60 min; the node space is contributors only (pages never become nodes), so both endpoints carry a lifecycle. Per-pair repeat counts (1,383 inductive test pairs) are min 1 / median 2 / mean 3.86 / p90 7 / max 156: CoEdit is recurrence-dominated, which is why the lifecycle drivers are informative (§7).

Protocol. 70/15/15 chronological split; inductive nodes are test nodes unseen in train/validation, inductive edges touch one. Negatives from a leak-audited pool (transductive seen-to-seen, inductive unseen-node), so a positive is never scored against an impossible negative. All stats strictly pre-update ($j < i$). All results mean $\pm$ sample std ($n - 1$) over seeds {1,7,42}; interventions over $N = 12{,}000$ pairs/seed.

Model and training. Hidden 128 (per-node GRU memory tied via `GRUCell(hidden,hidden)`, one knob). Adam (β=(0.9, 0.999), ε=10^{-8}), lr 10^{-3}, weight decay 10^{-5}, cosine annealing $T_{\max}$=20, 20 epochs, batch 500, **no dropout** (p=0). Transition loss weight 0.5; KL weighted λ_{KL}=10^{-2} on a term internally scaled by β=10^{-3} (effective 10^{-5}). Readout: ~150 learnable residuals over fixed priors; admissibility $C_{\text{BAND-5}}$ fixed; forbidden transitions suppressed ~20×. AP (where reported, §3.2/§3.3) is a model-agnostic average precision over the leak-audited pool; this paper reports no accuracy comparison: AP appears only as the quantity the readout does *not* move.

Hardware. Single GPU: RTX 3060; training ≈ 930–960 s (~16 min) per seed.

APPENDIX B: ARCHITECTURE SCHEMATICS

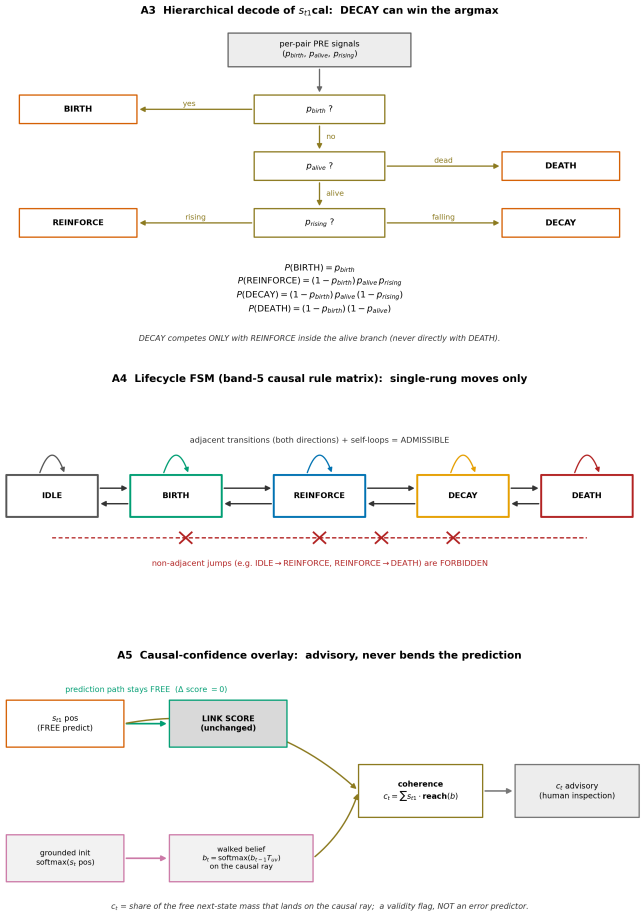


Figure B1. (a) Decode tree: gates $p_{\text{birth}}/p_{\text{alive}}/p_{\text{rising}}$ factor the five states (§3.1). (b) Admissibility band $C_{\text{BAND-5}}$: only adjacent transitions along the IDLE-BIRTH-REINFORCE-DECAY-DEATH axis permitted; IDLE→DEATH band-blocked (§3.3). (c) Free next-state prediction overlaid against the walked belief b_t; their agreement is c_t. Off by default, byte-identical when off (§5).